



Ghulam Ishaq Khan Institute of Engineering Sciences and Technology  
Faculty of Computer Science & Engineering

# MicroLend

Community Microfinance Loan Servicing and Delinquency Engine

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## SE431 — Software Quality Assurance

Milestone 1 — Quality Requirements, RTM and SQA Plan

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### Submitted by

|                  |         |
|------------------|---------|
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## Contents

|                                                                                  |           |
|----------------------------------------------------------------------------------|-----------|
| MicroLend — Community Microfinance Loan Servicing & Delinquency Engine . . . . . | 1         |
| 1. Quality Requirements (ISO/IEC 25010) . . . . .                                | 1         |
| 2. Requirements Traceability Matrix — Version 1 . . . . .                        | 3         |
| 3. SQA Plan — Scope and Outline (IEEE 730) . . . . .                             | 6         |
| 4. Cost of Quality and Quality-Culture Statement . . . . .                       | 8         |
| 5. Standards Awareness . . . . .                                                 | 10        |
| 6. Initial Risk Register . . . . .                                               | 10        |
| 7. Summary of M1 Quality Position . . . . .                                      | 14        |
| <b>Annex A — AI Usage Log</b>                                                    | <b>15</b> |
| Milestone 1 . . . . .                                                            | 15        |
| Milestone 2 . . . . .                                                            | 17        |
| Milestone 3 . . . . .                                                            | 17        |
| Rules the team has agreed for AI use on this project . . . . .                   | 17        |

## MicroLend — Community Microfinance Loan Servicing & Delinquency Engine

### Team

| Member           | Reg. #  | M1 SQA deliverable lead                      | Construction modules     |
|------------------|---------|----------------------------------------------|--------------------------|
| Muhammad Ibrahim | 2023446 | SQAP scope & outline, risk register          | origination, delinquency |
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**Repository:** <https://github.com/tughrul1/microlend> **Companion document:** docs/M1-SE423-Proposal-00D-Pattern-Plan.md — the construction-side proposal this quality plan applies to **Milestone:** M1 (Week 6) · **Weight:** 15% of the project grade

## 1. Quality Requirements (ISO/IEC 25010)

### 1.1 Which characteristics matter for this system, and which do not

ISO/IEC 25010 defines eight product quality characteristics. A quality plan that claims all eight matter equally has not analysed the system. MicroLend's product risk is **financial, not safety-related**, and it is an internal back-office tool with a handful of users, which determines where quality effort is spent:

| Characteristic                | Relevance to MicroLend | Why                                                                                                                                                 |
|-------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Functional suitability</b> | <b>Critical</b>        | A schedule or allocation that is arithmetically wrong corrupts every downstream figure — balances, penalties, classification, provisioning, reports |
| <b>Reliability</b>            | <b>Critical</b>        | A failure part-way through posting must not leave a half-written ledger; the end-of-day engine must survive a task dying                            |
| <b>Security (integrity)</b>   | <b>High</b>            | A ledger that can be altered is not an audit record. Integrity, not confidentiality, is the sub-characteristic that matters                         |
| <b>Maintainability</b>        | <b>High</b>            | The system is assessed across three milestones by three developers who must each work in code the others wrote                                      |
| <b>Performance efficiency</b> | <b>Moderate</b>        | The end-of-day run must finish within an overnight window. Nothing else is time-critical                                                            |

| Characteristic       | Relevance to MicroLend | Why                                                                                                                                |
|----------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usability</b>     | <b>Low</b>             | A small internal web interface used by trained branch staff on a local machine. Learnability is not a project risk                 |
| <b>Compatibility</b> | <b>Not applicable</b>  | A single self-contained application; no co-existence or interoperability requirement                                               |
| <b>Portability</b>   | <b>Not applicable</b>  | Runs locally on the JVM with a file-based database and is never deployed, so there is no deployment-environment variation in scope |

Compatibility and portability are declared **not applicable with justification** rather than quietly omitted.

## 1.2 The quality requirements

Each requirement carries exactly one measurable claim and names one of the four verification methods — **inspection, analysis, demonstration, test**.

| ID           | Requirement                                                                                                       | Characteristic         | Sub-characteristic            | Measurable target                                                                                                                             | Verification method and evidence                                                                                                                                          |
|--------------|-------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QR-01</b> | End-of-day processing of 10,000 active loans shall complete within 20 seconds on the demo machine using 4 threads | Performance efficiency | Time behaviour                | Median wall-clock over 5 consecutive runs $\leq 20$ s; no single run $> 25$ s                                                                 | <b>Test</b> — timed JUnit harness over a generated 10,000-loan dataset; timings attached to each milestone report                                                         |
| <b>QR-02</b> | Repayment allocation shall reproduce the reference schedules exactly, at the stated monetary scale                | Functional suitability | Functional correctness        | Zero discrepancy against 30 hand-computed golden schedules at scale 2, HALF_EVEN; 100% decision coverage of RepaymentAllocationLocator.post() | <b>Test</b> — golden-file unit tests plus a JaCoCo branch-coverage report                                                                                                 |
| <b>QR-03</b> | Any failure during repayment posting shall leave the ledger balanced, with no partial entry                       | Reliability            | Fault tolerance               | For 100% of at least 10 injected failure points: $\sum \text{debits} = \sum \text{credits}$ , and zero orphan entries                         | <b>Test</b> — fault injection through a repository stub that throws at each write point; also <b>demonstrated</b> live at M3 as one of the two required failure scenarios |
| <b>QR-04</b> | Core business-logic methods shall remain within the stated complexity and size limits                             | Maintainability        | Analysability / Modifiability | Mean cyclomatic complexity $\leq 5$ ; no method above 15; no class above 200 LOC, measured across the five module packages                    | <b>Analysis</b> — static-analysis metric report produced in CI and attached to each milestone                                                                             |

| ID           | Requirement                                                                                                            | Characteristic | Sub-characteristic | Measurable target                                                                                                                                                                                  | Verification method and evidence                                                                                                                               |
|--------------|------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QR-05</b> | A posted ledger entry shall not be alterable by any component; corrections shall occur only as compensating entries    | Security       | Integrity          | Zero mutating members on <code>LedgerEntry</code> ; zero <code>UPDATE</code> statements against <code>ledger_entry</code> in the codebase; 100% of corrections implemented as compensating entries | <b>Inspection</b> — an explicit item on the Fagan inspection checklist, supported by a static-analysis rule and a unit test asserting reversal-only correction |
| <b>QR-06</b> | The end-of-day engine shall produce identical results whether or not repayments are posted concurrently during the run | Reliability    | Maturity           | Over 50 repeated runs of the concurrency suite: zero lost updates, zero double accruals, identical final balances to the single-threaded reference run                                             | <b>Test</b> — deterministic latch-driven concurrency tests; the suite is repeated 50 times nightly in CI                                                       |

### 1.3 Constraints (not quality requirements)

The standard separates requirements into functional, non-functional (quality) and **constraints**. The following are constraints on how the system is built. They are recorded in SQAP §4.4 rather than presented as quality requirements:

| ID          | Constraint                                                                                                        | Rationale                                                                                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C-01</b> | Monetary values shall never be represented as double or float                                                     | $0.1 + 0.2 \neq 0.3$ in IEEE 754; a fraction of a rupee per instalment accumulates into a ledger that does not balance. Enforced by a Checkstyle rule that fails the build |
| <b>C-02</b> | <code>BigDecimal</code> values shall be compared with <code>compareTo() == 0</code> , never <code>equals()</code> | <code>equals()</code> also compares scale, so <code>2.50</code> is not equal to <code>2.5</code>                                                                           |
| <b>C-03</b> | No cyclic dependencies between module packages                                                                    | Cycles destroy the ability to test or replace a module in isolation                                                                                                        |
| <b>C-04</b> | Monetary columns shall be stored as integer minor units                                                           | Same reasoning as C-01, applied at the storage layer                                                                                                                       |

### 1.4 The quality trade-off

**QR-01 (performance) is in tension with QR-03, QR-05 and QR-06 (fault tolerance, integrity, maturity).** Per-account locking, immutable value objects, append-only correction and an idempotency key all cost time behaviour: each adds work on the path that QR-01 measures.

The conflict is resolved by **capping the cost rather than removing either attribute** — locks are per account rather than global, so unrelated accounts proceed in parallel and the lock is never held across I/O. The measured cost is recorded in the GQM data at each milestone rather than asserted to be small.

## 2. Requirements Traceability Matrix — Version 1

### 2.1 Purpose and how it will be maintained

The RTM supports traceability in **both directions**. Forward — from requirement to code to test — answers “*did we build and verify everything that was asked for?*” and exposes a **test gap**: a requirement implemented but never verified. Backward — from code to requirement — answers “*why does this code exist?*” and exposes **rogue code**: a module traceable to no approved requirement.

Those are the two defects this matrix is audited for at every milestone. At M1 the design and class columns are planned rather than built, so they are marked *planned*; from M2 they name real artifacts and real test results.

The matrix is maintained at `qa/rtm.md` in the repository and versioned with the code, so the state of traceability at any baseline can be reproduced from the tag.

## 2.2 Functional requirements

| ID    | Requirement                                                                                             | Module      | Design element (planned)                               | Test approach   | Test design technique                                                                   |
|-------|---------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------|
| FR-01 | An applicant shall be at least 18 years old at disbursement and no more than 65 at maturity             | origination | AgeRule                                                | Unit            | <b>BVA:</b> 17 / 18 / 65 / 66 at both dates                                             |
| FR-02 | An applicant shall hold no more than 2 active loans                                                     | origination | ActiveLoanLimitRule                                    | Unit            | <b>EP + BVA:</b> 0, 1, 2, 3 active loans                                                |
| FR-03 | Total monthly instalment burden shall not exceed 40% of declared monthly income                         | origination | BurdenRatioRule                                        | Unit            | <b>BVA:</b> 39.99% / 40.00% / 40.01%                                                    |
| FR-04 | An applicant with a prior write-off shall be rejected                                                   | origination | WriteOffHistoryRule                                    | Unit            | <b>EP:</b> no history / written-off / settled late                                      |
| FR-05 | A rejected application shall state a typed rejection reason, not a boolean                              | origination | Decision, RejectionReason                              | Unit            | <b>EP:</b> one case per rejection reason                                                |
| FR-06 | Risk grade A/B/C shall adjust the rate by a fixed spread and cap the approved amount                    | origination | RiskGrade, Decision                                    | Unit            | <b>EP:</b> one case per grade                                                           |
| FR-07 | The system shall generate schedules for flat-rate, reducing-balance and bullet products                 | products    | FlatRateProduct, ReducingBalanceProduct, BulletProduct | Unit            | <b>Golden file:</b> 10 hand-computed schedules per product                              |
| FR-08 | Instalment dates shall follow the disbursement day-of-month, clamped for short months                   | products    | DueDateCalculator                                      | Unit            | <b>BVA:</b> disbursement on 28th, 29th, 30th, 31st; February in leap and non-leap years |
| FR-09 | The instalments of a schedule shall sum exactly to principal plus total interest                        | products    | ScheduleAssembler                                      | Unit (property) | <b>Property-based:</b> for all valid amount/tenure/rate combinations                    |
| FR-10 | A grace period shall defer the first instalment while still accruing interest                           | products    | GracePeriodPolicy                                      | Unit            | <b>EP + BVA:</b> grace of 0 / 1 / 3 months                                              |
| FR-11 | Repayments shall be applied in the order penalty → fees → interest → principal, oldest instalment first | ledger      | RepaymentAllocator, <del>Interest</del>                | Unit            | <b>Basis path</b> (CC ≈ 11) + EP on payment size                                        |

| ID    | Requirement                                                                                                     | Module      | Design element (planned)              | Test approach      | Test design technique                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------|--------------------|---------------------------------------------------------------------------------|
| FR-12 | A partial payment shall be applied as far as it reaches, leaving the remainder outstanding                      | ledger      | RepaymentAllocator                    | Unit               | <b>BVA:</b> 0 / 0.01 / due – 0.01 / exactly due                                 |
| FR-13 | An overpayment shall become an advance credit unless it clears the loan, in which case early settlement applies | ledger      | RepaymentAllocator, SettlementPolicy  | Unit               | <b>BVA:</b> due + 0.01 / exact payoff / payoff + 0.01                           |
| FR-14 | A posted ledger entry shall be immutable; corrections occur only as compensating entries                        | ledger      | LedgerEntry, CompensatingEntryService | Unit + inspection  | <b>Inspection checklist item</b> + unit test asserting no mutator exists        |
| FR-15 | Every posting shall write balanced debit and credit lines                                                       | ledger      | CoreRepaymentPosting                  | Unit (invariant)   | <b>Invariant test:</b> $\Sigma$ debits = $\Sigma$ credits after every posting   |
| FR-16 | Penalty shall accrue at 0.05% per day on overdue principal, capped at 10% of the overdue amount                 | delinquency | PenaltyAccruer                        | Unit               | <b>BVA:</b> at the cap boundary; EP on days overdue                             |
| FR-17 | Accounts shall be bucketed 0 / 1–30 / 31–60 / 61–90 / 90+ days past due and classified accordingly              | delinquency | DelinquencyClassifier                 | Unit               | <b>Basis path</b> (CC $\approx$ 8) + <b>BVA:</b> 0/1/30/31/60/61/90/91          |
| FR-18 | A loan more than 180 days past due shall be written off automatically                                           | delinquency | WriteOffPolicy                        | Unit               | <b>BVA:</b> 179 / 180 / 181 days                                                |
| FR-19 | A provision percentage shall be applied per classification                                                      | delinquency | ProvisioningPolicy                    | Unit               | <b>EP:</b> one case per classification                                          |
| FR-20 | The system shall produce a customer statement for a date range                                                  | reporting   | StatementRenderer                     | Unit               | <b>EP:</b> empty range / single entry / full history                            |
| FR-21 | The system shall report portfolio at risk over 30 days, provision coverage and collection efficiency            | reporting   | PortfolioReporter                     | Unit               | <b>Golden file:</b> a fixed portfolio with hand-computed metrics                |
| FR-22 | Repayments shall be importable from bank statement files in each supported bank's format                        | ledger      | CsvStatementAdapter                   | Unit + integration | <b>EP:</b> valid row / malformed date / minor-unit amount / duplicate reference |

## 2.3 Quality requirements traced

| ID    | Verification method  | Primary artifact (planned)                                     | Evidence produced at |
|-------|----------------------|----------------------------------------------------------------|----------------------|
| QR-01 | Test                 | EndOfDayPerformanceTest                                        | M2, M3               |
| QR-02 | Test                 | GoldenScheduleTest,<br>JaCoCo branch report                    | M2, M3               |
| QR-03 | Test + demonstration | PostingFaultInjectionTest                                      | M2, demonstrated M3  |
| QR-04 | Analysis             | Static-analysis metric report in<br>CI                         | M2, M3               |
| QR-05 | Inspection           | Fagan inspection checklist<br>item +<br>LedgerImmutabilityTest | M2                   |
| QR-06 | Test                 | EndOfDayConcurrencyTest,<br>50× nightly                        | M2, M3               |

## 2.4 Coverage of the matrix at M1

| Measure                                           | Status at M1                   |
|---------------------------------------------------|--------------------------------|
| Functional requirements defined                   | 22                             |
| Quality requirements defined                      | 6                              |
| Constraints recorded                              | 4                              |
| Requirements with a planned verification approach | 28 of 28 (100%)                |
| Requirements with executed tests                  | 0 — M1 is a planning milestone |
| Code modules traceable to a requirement           | n/a at M1; audited from M2     |

**Audit rule for M2 and M3:** every requirement must have at least one test (no test gaps), and every production class must trace to at least one requirement (no rogue code). Both directions are checked, because neither direction can find the other's defect.

## 3. SQA Plan — Scope and Outline (IEEE 730)

The outline below follows the IEEE 730-2014 structure. Every mandatory section is listed with a one-line plan specific to MicroLend. The full plan is developed across M2 and completed at M3; M1 establishes the scope and the structure.

| §   | IEEE 730 section              | Plan for MicroLend                                                                                                                                                                                                                                                                                     |
|-----|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Purpose and scope             | This SQAP governs the MicroLend loan-servicing system across M1–M3, covering the five modules and the shared repository, tailored to a three-person Very Small Entity                                                                                                                                  |
| 2   | Definitions and acronyms      | Define the terms actually used — DPD, PAR30, allocation waterfall, EOD, compensating entry, cyclomatic complexity, EP/BVA — plus error, defect and failure, so the document is self-contained                                                                                                          |
| 3   | Reference documents           | The joint-project guidelines, ISO/IEC 25010, ISO/IEC/IEEE 29148, IEEE 730-2014, IEEE 1028, ISO/IEC 29110, the Business Rules Register and the RTM                                                                                                                                                      |
| 4.1 | Organisation and independence | Name the three members, their module ownership and the rotating SQA lead. State honestly that organisational independence is impossible in a three-person team, and that it is substituted by <b>non-owner review</b> (no one approves their own pull request) and the <b>external inspector at M2</b> |

| §     | IEEE 730 section                                    | Plan for MicroLend                                                                                                                                                                                                                                            |
|-------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.2   | Software product risk                               | Classify MicroLend's product risk as <b>financial, not safety-related</b> ; identify the money path (RepaymentAllocator, DelinquencyClassifier, the ledger) as the high-risk area; set SQA depth from that, since SQA effort is proportionate to product risk |
| 4.3   | Tools                                               | Java and JUnit 5, SQLite, Gradle, GitHub and GitHub Actions, JaCoCo for coverage, Checkstyle and a static analyser — each named with the quality question it answers                                                                                          |
| 4.4   | Standards, practices and conventions                | Constraints C-01 to C-04; the commit convention <code>test(red): / feat(green): / refactor::</code> ; the documentation standard; and the <b>100% decision coverage on the money path</b> policy                                                              |
| 4.5   | Effort, resources and schedule                      | Approximately 39 student-days of construction plus about 8 student-days of SQA work per milestone, mapped onto the Week 6 / 10 / 13 dates with named owners                                                                                                   |
| 5.1.1 | Evaluate plans for conformance                      | At each milestone, check that the M1 proposal, the test plan and this SQAP still describe what is actually being built; record deviations                                                                                                                     |
| 5.1.2 | Evaluate product for conformance                    | The Fagan inspection of the two money-path classes, static-analysis triage, and black-box and white-box test execution against the RTM requirement IDs                                                                                                        |
| 5.1.3 | Evaluate product for acceptability                  | Define the M3 acceptance gate: all functional requirements traced to passing tests, $\geq 60\%$ coverage of core logic with 100% decision coverage on the money path, zero open critical defects                                                              |
| 5.1.4 | Evaluate product life-cycle support for conformance | Check at each baseline that the README, setup instructions, schema DDL and design documents match the code                                                                                                                                                    |
| 5.1.5 | Measure products                                    | Defect density per module, branch coverage, cyclomatic complexity per method, open and closed defect counts — the metrics the GQM questions consume                                                                                                           |
| 5.2.1 | Evaluate life-cycle processes for conformance       | Verify that the red-green-refactor sequence, pull-request review and merge gates were genuinely followed, using commit and pull-request history as the evidence                                                                                               |
| 5.2.2 | Evaluate environments for conformance               | Confirm all three members build and test on the same toolchain versions, and that CI is the single source of truth for a green build                                                                                                                          |
| 5.2.3 | Evaluate subcontractor processes                    | <b>Declared not applicable, with justification:</b> no subcontractors and no third-party components beyond the JDBC driver and test libraries. The standard requires the declaration, not silence                                                             |
| 5.2.4 | Measure processes                                   | Inspection rate in LOC per hour, defect-removal effectiveness by phase, and rework as a percentage of total effort                                                                                                                                            |

| §     | IEEE 730 section                     | Plan for MicroLend                                                                                                                                                                                                      |
|-------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.2.5 | Assess staff skill and knowledge     | Record that concurrency, white-box test design, static analysis and the Fagan inspection process are <b>not yet taught</b> at M1, and name the self-study and pairing that closes each gap before the technique is used |
| 6.1   | Contract review                      | Treat the joint-project guidelines as the agreement: a checklist mapping every Section 3.1 and 3.2 minimum to where it is met, re-checked at each milestone                                                             |
| 6.2   | Quality measurement                  | The GQM programme — one goal, three questions, and the metrics listed in §5.1.5 and §5.2.4                                                                                                                              |
| 6.3   | Waivers and deviations               | Any departure from this plan is logged with reason, approver and date — for example, dropping a planned pattern is recorded, not silently abandoned                                                                     |
| 6.4   | Task repetition                      | A re-inspection is triggered if an inspection finds more than a set number of major defects; the full regression suite re-runs on every push                                                                            |
| 6.5   | Risk to performing SQA               | The honest risks to the quality work itself — exam weeks compressing the inspection, the external inspector falling through, one member carrying the SQA load                                                           |
| 6.6   | Communications strategy              | A weekly 15-minute session where each member explains their module's invariants to the other two; defect log and corrective actions in the repository; milestone status summarised in the M2 and M3 documents           |
| 6.7   | Non-conformance process              | How a defect or process non-conformance is raised, triaged by severity, assigned, corrected and verified — the corrective-action loop closed at M2 and M3                                                               |
| 7.1   | SQA records: collect, file, maintain | All quality records live in /qa in the repository: inspection log, defect log with corrective-action status, static-analysis reports, coverage reports, GQM data and risk-register versions                             |
| 7.2   | Availability of records              | Records are versioned with the code in the hosted repository, so any record can be produced on request at the demo                                                                                                      |

### 3.1 Inspection planning note (for M2)

The Fagan inspection process runs in six stages: **plan** → **kickoff** → **individual document checking** → **logging meeting** → **edit** → **follow-up and exit**. The planning guideline is **100–200 lines of code per hour** for source code.

That rate is the number to plan with. Inspecting `RepaymentAllocator` plus `DelinquencyClassifier` — roughly 250 lines — means a logging meeting of about 1.5 to 2 hours, which is schedulable with an external inspector and which also provides the denominator for the defect-detection-rate metric in the GQM programme. A team that ignores the rate “inspects” 800 lines in forty minutes and finds nothing, and the defect log then shows exactly that.

## 4. Cost of Quality and Quality-Culture Statement

### 4.1 The model

Total cost of quality is the sum of the **cost of conformance** (what is spent to prevent and detect defects) and the **cost of non-conformance** (what defects cost once they exist).

## 4.2 Expected costs for this project, in student-hours

The currency here is time, because that is the resource the team actually spends. Figures are estimates made at M1 and compared against actuals at M3 — the comparison is itself a GQM metric.

| Category                | Activity for MicroLend                                                                   | Estimated effort (student-hours)          |
|-------------------------|------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Prevention</b>       | Writing the Business Rules Register and freezing scope                                   | 6                                         |
|                         | Coding standard and Checkstyle configuration (constraints C-01 to C-04)                  | 4                                         |
|                         | Hand-computing 30 golden schedules in a spreadsheet before coding                        | 8                                         |
|                         | Design review of the class model before implementation                                   | 6                                         |
|                         | Setting up CI, branch protection and quality gates                                       | 5                                         |
|                         | <b>Prevention subtotal</b>                                                               | <b>29</b>                                 |
| <b>Appraisal</b>        | Writing and maintaining the test suite (test-first)                                      | 60                                        |
|                         | Pull-request review by a non-owner                                                       | 18                                        |
|                         | Fagan inspection of the money path, including preparation                                | 12                                        |
|                         | Static-analysis runs and triage                                                          | 8                                         |
|                         | <b>Appraisal subtotal</b>                                                                | <b>98</b>                                 |
| <b>Internal failure</b> | Rework of defects found by tests, review or inspection before submission                 | 25 (expected)                             |
| <b>External failure</b> | Rework of defects found by the grader at a milestone demo, or by the following milestone | 10 (target: as close to zero as possible) |
|                         | <b>Non-conformance subtotal</b>                                                          | <b>35</b>                                 |
|                         | <b>Total expected cost of quality</b>                                                    | <b>162 student-hours</b>                  |

## 4.3 What the team expects these numbers to show

Three things follow from this table, and each is a prediction the team is prepared to be held to at M3:

1. **Appraisal dominates, and that is correct for this project.** The money path has no safety net other than tests: a wrong instalment does not crash, it quietly produces a wrong number. Detection effort is where the risk is.
2. **Internal failure cost rising is a good sign, not a bad one.** If the register shows more hours of rework triggered by our own tests and inspections at M2 than at M1, it means defects are being caught inside the team rather than at a demo. The number to minimise is **external** failure, not internal. A team reporting zero internal failure cost is reporting that it is not looking.
3. **The 1:10:100 principle is why the 8 hours on golden schedules are spent first.** A rounding rule corrected while writing the spreadsheet costs minutes. The same defect found after schedules, ledger entries, delinquency buckets and portfolio reports have all been built on it requires reworking every one of them, under deadline. Prevention is bought at the cheapest point on the curve.

## 4.4 Quality-culture statement

The team commits to four behaviours, each with a mechanism that makes it checkable rather than aspirational:

| Commitment                                                                                                                        | The mechanism that enforces it                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Quality is not a phase at the end.</b> Quality activities run in the same milestone cycle as the construction work they assess | SQA deliverables are owned per milestone and rotate; the M2 inspection targets M2 code, not a frozen artifact                                        |
| <b>No one marks their own homework.</b> Every change is reviewed by someone who does not own that module                          | Branch protection requires one approving review from a non-author; CODEOWNERS routes the request                                                     |
| <b>Defects are data, not blame.</b> A defect found in inspection is a success for the process                                     | The defect log records type, severity and phase found — never who wrote it. The metric is defect-removal effectiveness, not individual defect counts |
| <b>If it is not evidenced, it did not happen.</b> Every claim in this plan maps to an artifact in the repository                  | All quality records live in /qa, versioned with the code and reproducible from a milestone tag                                                       |

The honest limit: in a three-person team, the person reviewing is also a developer on the project. Independence is structural, not organisational, and that is stated in SQAP §4.1 rather than hidden.

## 5. Standards Awareness

Software has no laws of nature to stop a defect from executing. A silent rounding error or a lost update in our ledger meets no physical resistance the way an under-designed arch meets gravity, so process standards serve as the surrogate laws — and they are a **prevention cost** in the cost-of-quality model rather than paperwork. By business model, MicroLend is *custom software written in-house*: the category where standards are adopted voluntarily to stop chronic rework rather than imposed by a client.

**ISO/IEC/IEEE 12207** tells us *what processes must exist*. Of its supporting processes we genuinely perform Configuration Management (branch protection, the m1/m2/m3 baseline tags), Verification and Validation, Joint Review and Quality Assurance. **IEEE 730-2014** tells us *how* the SQA function executes inside that lifecycle: its activities in three groups — SQA process implementation, product assurance, process assurance — its principle that the depth of SQA is set by **product risk**, and its rule that any activity not performed must be declared *not applicable with a justification* rather than quietly dropped. That rule is why our SQAP §5.2.3 explicitly marks subcontractor evaluation as not applicable instead of omitting it.

Full 12207 and CMMI adoption would be wrong at our scale. Three students on one project is a **Very Small Entity** under ISO/IEC 29110's 25-person definition, and at roughly two person-months of effort we fall formally in the **Entry** profile, though we follow the **Basic** profile's structure as taught — its two processes, Project Management and Software Implementation, map one-to-one onto our three milestones, and its Deployment Packages supply ready-made templates and traceability tables so our time goes into the work rather than into inventing document formats. Lightweight here means *scaled to context*, not undisciplined.

Against **CMMI-DEV** we claim **Maturity Level 2 behaviours only**, and we name them: Requirements Management through the RTM, Configuration Management through protected branches and baselines, Process and Product Quality Assurance through the Fagan inspection and static-analysis triage, and Measurement and Analysis through the GQM programme. We explicitly do **not** claim Level 3, because a maturity level describes an organisation's standard processes rather than a single project's, and the statistical process control of Levels 4 and 5 is meaningless with three people and one data point per milestone.

From **ISO 9001:2015** we take two things concretely. The **PDCA cycle** is literally our milestone loop: plan the milestone, build it, measure coverage and defect density, then change the *process* rather than promising to be more careful. **Risk-based thinking** is why our register already records that only one member fully understands the allocation waterfall, although nothing has failed yet. **ISO/IEC 90003** is what licenses us to translate ISO 9001's abstract "design and development verification" into the specific software acts we actually perform — code inspection, static analysis, automated unit tests and decision-coverage criteria.

Finally, rigour is scaled to consequence. MicroLend is not safety-critical, so the domain standards for airborne systems, railway applications and medical devices do not apply, and we make no such claim. But because it is a financial ledger we borrow their one transferable idea — that criticality dictates technique — and adopt as a stated SQAP policy **100% decision coverage on the money path** (RepaymentAllocator, DelinquencyClassifier), together with the coding-standard rule banning double for monetary values: our scaled-down equivalent of a safety standard's language restrictions, recorded under IEEE 730's "standards, practices and conventions" clause rather than left as an aspiration.

**What we are not claiming**, stated plainly: CMMI Level 3 or above; conformance to any safety-critical standard; organisational independence of the QA function; and certification of any kind. Every standard named above is used as a source of practice, not as a badge.

## 6. Initial Risk Register

The full register is maintained as a living document at `qa/risk-register.md` in the repository, updated at every milestone and whenever a risk materialises or closes. It is reproduced here as at M1.

### 6.1 Scoring

Risk exposure is **computed, not asserted**:  $E = P \times C$ , where  $P$  is probability (1 = rare ... 5 = almost certain) and  $C$  is consequence (1 = negligible ... 5 = severe).

| Exposure | Priority | Response                                      |
|----------|----------|-----------------------------------------------|
| 17–25    | Critical | Act now; mitigation owned and tracked weekly  |
| 10–16    | High     | Mitigation planned and started this milestone |
| 5–9      | Medium   | Monitor; mitigate if exposure rises           |
| 1–4      | Low      | Accept; review each milestone                 |

## 6.2 The register at M1

| ID        | Risk<br>(condition<br>→ consequence)                                                                                                                        | Type    | P | C | E         | Priority | Mitigation                                                                                                                                                                                            | Owner           | Status |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|-----------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|
| <b>R1</b> | Money or date arithmetic in schedule generation or the allocation waterfall is subtly wrong → every downstream figure is wrong and the defect is found late | Product | 4 | 5 | <b>20</b> | Critical | Hand-compute 30 golden schedules before coding and commit them as the first tests; BigDecimal only, enforced by constraint C-01; property test that $\Sigma$ instalments = principal + total interest | Hassan Khalid   | Open   |
| <b>R2</b> | The team starts M1 late → the compressed schedule pushes work into the M2 window, and M2 carries 35%                                                        | Project | 5 | 3 | <b>15</b> | High     | Freeze scope at five modules with no new features; draft M1 documents from the existing analysis; set up repository and tooling in parallel with document writing                                     | Tughrul Hussain | Open   |

| ID        | Risk<br>(condition<br>→ consequence)                                                                                                           | Type    | P | C | E         | Priority | Mitigation                                                                                                                                                                         | Owner            | Status |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------|
| <b>R3</b> | The external inspector required at M2 is not arranged in time → the inspection criterion (20% of SE431 M2) cannot be fully evidenced           | Process | 3 | 4 | <b>12</b> | High     | Request the instructor's team pairing immediately after M1; agree a date in Week 8; prepare the inspection package in advance                                                      | Muhammad Ibrahim | Open   |
| <b>R4</b> | Commits are squashed or tests are written after the code → the red-green-refactor trail cannot be reconstructed, losing TDD marks at M2 and M3 | Process | 3 | 4 | <b>12</b> | High     | Squash-merge disabled in repository settings from day one; commit convention enforced; merges use --no-ff; the pull-request template asks for the red commit hash                  | Tughrul Hussain  | Open   |
| <b>R5</b> | Concurrency defects in the end-of-day engine appear only under load → late, hard-to-reproduce failures near M3                                 | Product | 3 | 4 | <b>12</b> | High     | Immutable value objects; per-account striped locks with ascending lock ordering; a unique (loan_id, business_date) idempotency key; deterministic latch-driven tests with no sleep | Muhammad Ibrahim | Open   |

| ID        | Risk<br>(condition<br>→ consequence)                                                                                                         | Type    | P | C | E         | Priority | Mitigation                                                                                                                                                                                  | Owner            | Status |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|-----------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------|
| <b>R6</b> | Individual contribution is not attributable → individual evidence suffers, which is 20% of each member's milestone mark                      | Project | 3 | 3 | <b>9</b>  | Medium   | Module ownership equals directory ownership; every change by pull request reviewed by a non-owner; SE431 deliverable ownership rotates each milestone; AI Usage Log maintained continuously | Tughrul Hussain  | Open   |
| <b>R7</b> | Several M2-graded techniques are not taught until after they must be designed for → the team designs ahead of the lectures and gets it wrong | Project | 4 | 4 | <b>16</b> | High     | Design the structures now, defer detailed technique work until taught; start the four non-reconstructable evidence streams at M1; re-check each design against the lecture when delivered   | Muhammad Ibrahim | Open   |

| ID        | Risk<br>(condition<br>→ conse-<br>quence)                                                                                                   | Type    | P | C | E         | Priority | Mitigation                                                                                                                                                | Owner         | Status |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|-----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|
| <b>R8</b> | AI-assisted work is submitted that a member cannot explain in the viva → reduced or zero individual marks, and a possible integrity concern | Project | 3 | 5 | <b>15</b> | High     | AI Usage Log updated continuously; the module owner reviews and rewrites generated code in their own module; dry-run Q&A inside the team before each demo | Hassan Khalid | Open   |

### 6.3 Review schedule

- **M1:** register initiated with eight risks, scored and owned.
- **M2:** every risk re-scored using evidence from the architecture, inspection, static analysis, testing and SCM; risks arising from implementation added; residual exposure recorded.
- **M3:** continuous updating from M1 to M3 demonstrated; mitigation effectiveness evaluated against actual project data; residual risk recorded.

## 7. Summary of M1 Quality Position

| Deliverable          | Status at M1                                                                                                                               |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Quality requirements | 6 defined, each with a measurable target and a named verification method; 2 characteristics declared not applicable with justification     |
| Constraints          | 4 recorded, C-01 enforced by the build                                                                                                     |
| RTM v1               | 22 functional and 6 quality requirements, each with a planned verification approach and a named test design technique                      |
| SQAP                 | Scope set and all mandatory IEEE 730 sections outlined with project-specific plans; one section declared not applicable with justification |
| Cost of Quality      | Estimated at 162 student-hours with the prevention/appraisal/failure split and three stated predictions                                    |
| Standards awareness  | 12207, IEEE 730, 29110, CMMI-DEV, ISO 9001 and 90003 related to this project's scale, with explicit non-claims                             |
| Risk register        | 8 risks, computed exposure, owners assigned, living document in the repository                                                             |

**What M1 deliberately does not yet contain**, because the techniques are taught later in the semester and the rubric does not require them at this milestone: executed test results, the inspection defect log, the static-analysis report, coverage figures and GQM data. Each has a planned home in this document and an owner, and the four evidence streams that cannot be reconstructed retrospectively — the TDD commit trail, SCM baselines, the static-analysis baseline and GQM data collection — have been started at M1 even though they are assessed later.

## Annex A — AI Usage Log

**Required by the joint-project guidelines §9 (Academic Integrity & AI Use), submitted with every milestone.**

The guidelines permit AI-assisted development, but assessment is based on whether the team understands, verifies and can defend what it submits. Any member who cannot explain or justify work claimed as theirs may receive reduced individual marks regardless of the team's score.

**How this log is maintained:** one row per significant use, added **at the time of use**, not reconstructed before a deadline. The *Verification* column names what was actually checked.

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### Milestone 1

#### Entry 1 — Select a project that satisfies both course rubrics

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked for candidate projects meeting every Section 3.1 and 3.2 minimum, simple to build, and defensible in a viva. Three candidates were analysed and scored against the rubric.

**Adopted.** The comparative analysis in `project/Project-Idea-Analysis.md`; MicroLend selected over SkyHold (air-line seat inventory) and DispenseRx (pharmacy dispensing).

**Modifications made.** Scope reduced to an application that runs locally with no deployment, no hosted services and no paid resources. DispenseRx rejected by the team on the stated ground that its clinical data would be invented and therefore indefensible under questioning.

**Verification performed.** Fit table checked row by row against the guidelines PDF, with each claimed minimum traced to where it is met. Team reviewed and accepted the recommendation on 4 October 2026.

#### Entry 2 — Draft the SE423 M1 document

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked for a document following the six M1 rubric criteria for the chosen project.

**Adopted.** Draft of `docs/M1-SE423-Proposal-00D-Pattern-Plan.md`: class model, SOLID analysis, pattern justifications and the cohesion/coupling reasoning.

**Modifications made.** The single wide class diagram was replaced by three per-area views after the first PDF proved illegible. “Command-line interface” was replaced throughout with a locally served web interface, once the team confirmed the intended form of the application.

**Verification performed.** Pattern choices checked against the taught material. An earlier draft's “Factory Method” was found to be a Simple Factory — which Lecture 11 states is *not* the GoF pattern — and was restructured so that `LoanProduct` subclasses are the `ConcreteCreators`. Team reviewed and accepted the document on 4 October 2026.

#### Entry 3 — Draft the SE431 M1 document

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked for quality requirements, an RTM, the SQAP outline, Cost of Quality, standards awareness and a risk register, against the SE431 M1 rubric.

**Adopted.** Draft of `docs/M1-SE431-Quality-Requirements-RTM-SQAP.md`.

**Modifications made.** Usability and portability entries rewritten for a locally-run web application. Quality requirements restated so each names one of the four standard verification methods, and four items reclassified from quality requirements to constraints (C-01 to C-04).

**Verification performed.** Requirement targets and the Cost of Quality estimate reviewed for realism against the team's available time. Team reviewed and accepted the document on 4 October 2026.

#### Entry 4 — Repository, quality gates and coding standard

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked to create the repository with branch protection, CODEOWNERS, a pull-request template, and a Checkstyle configuration enforcing the money constraint.

**Adopted.** `config/checkstyle.xml`, `.github/CODEOWNERS`, `.github/pull_request_template.md`, and the README branching section.

**Modifications made.** The C-01 check was rewritten from a text pattern to a token-based rule (`IllegalToken` on `NUM_DOUBLE` and `NUM_FLOAT`) after CI showed the original flagging the version string `"0.1.0-M1"` as a decimal literal.

**Verification performed.** Branch protection tested by attempting a direct push to `main`, which GitHub reported as a bypassed rule violation. The Checkstyle fault was found by the CI pipeline itself and corrected before the build went green.

#### Entry 5 — Business Rules Register

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked to turn the described rules into a numbered register with boundary values for test design.

**Adopted.** `docs/business-rules.md` — 43 rules across the five modules.

**Modifications made.** None to the rule set. The six numeric decisions listed in §8 — the 40% burden ceiling, the 0.05%/day penalty capped at 10%, the 180-day write-off, the risk-grade spreads, the provision percentages and the 3-month grace cap — were reviewed and **accepted unchanged** by the team on 4 October 2026.

**Verification performed.** Boundary values cross-checked against the test design techniques recorded in the RTM, so every rule carrying a threshold has a corresponding boundary test planned.

#### Entry 6 — Build, test scaffold and CI pipeline

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked for a Gradle build, the first domain class written test-first, and a CI pipeline, with nothing that incurs cost.

**Adopted.** `build.gradle`, the Gradle wrapper, `Money`, `LoanDomainException`, `InvalidMoneyException`, `MoneyTest`, and `.github/workflows/build.yml`.

**Modifications made.** Written as a genuine TDD cycle: the failing test was committed first (`test(red):`), then the implementation (`feat(green):`). `Money.minus` was given a negative-result guard so the type cannot express direction — the ledger's debit and credit columns do that.

**Verification performed.** CI run on the public repository: compile, 10 JUnit tests, the Checkstyle gate and the coverage report all pass. Two faults were found and fixed through CI — a missing executable bit on `gradlew`, and the Checkstyle false positive described in entry 4.

#### Entry 7 — Submission PDFs

**Date:** 4 October 2026 · **Tool:** Claude Opus 5 (Claude Code)

**Activity.** Asked for a PDF build with a GIKI title page carrying the team names, project title and course code.

**Adopted.** `docs/build-pdf.sh` and `docs/header.tex`.

**Modifications made.** Fonts changed to Times New Roman with explicit Unicode mappings.

**Verification performed.** Rendered pages inspected individually. A font fault was found that silently dropped the symbols for “less than or equal to”, “greater than or equal to”, “approximately” and “multiplied by” from the PDFs — which would have corrupted the stated quality targets — and was corrected before submission.

#### Declaration for Milestone 1

I confirm that I have read, understood and verified the material submitted under my name, that I can explain and defend the design decisions, quality requirements and analysis it contains, and that any AI-assisted content was reviewed and modified by me where needed.

**Muhammad Ibrahim — 2023446** Sections owned: origination, delinquency; SQAP outline, risk register

Signature: ..... Date: .....

**Hassan Khalid — 2023242** Sections owned: products, reporting; quality requirements, RTM

Signature: ..... Date: .....

**Tughral Hussain — 2023532** Sections owned: ledger; standards awareness, Cost of Quality, repository and build setup

Signature: ..... Date: .....

**Milestone 2**

*(rows added as work proceeds)*

| # | Date | Tool | Purpose | Prompt /<br>activity<br>summary | Adopted | Modifications | Verification |
|---|------|------|---------|---------------------------------|---------|---------------|--------------|
|---|------|------|---------|---------------------------------|---------|---------------|--------------|

**Milestone 3**

*(rows added as work proceeds)*

| # | Date | Tool | Purpose | Prompt /<br>activity<br>summary | Adopted | Modifications | Verification |
|---|------|------|---------|---------------------------------|---------|---------------|--------------|
|---|------|------|---------|---------------------------------|---------|---------------|--------------|

**Rules the team has agreed for AI use on this project**

1. **Generated code enters the repository only through the module owner**, who reviews it line by line and rewrites what they would have written differently. The owner is the one answering for it in the viva.
2. **No generated test is accepted without reading what it asserts.** A test that passes for the wrong reason is worse than no test.
3. **Business rules and numeric thresholds are the team’s decisions**, not suggestions accepted by default.
4. **Every row in this log is added when the work happens.** A log assembled the night before a deadline is itself evidence of the problem the requirement is designed to catch.
5. **If a member cannot explain a submitted artifact, it is withdrawn or rewritten before submission** rather than defended in the demo.